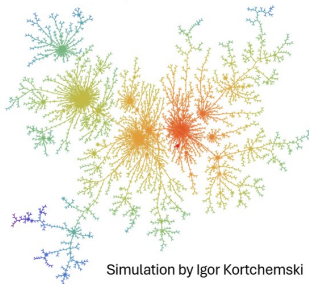


Mass erasure of infinitely measured Polish R -trees, and applications to Lévy forests

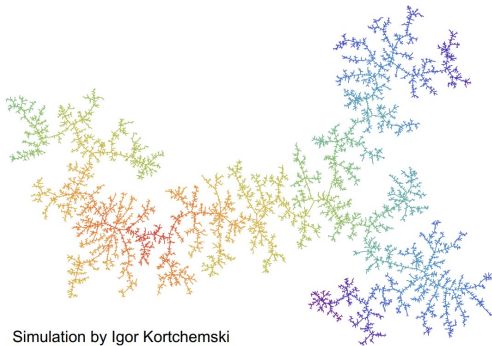
Discrete Random Structures 2026, Bedlewo

Mie Glückstad (supervised by Prof. Matthias Winkel)



Classical scaling limit: The Brownian CRT

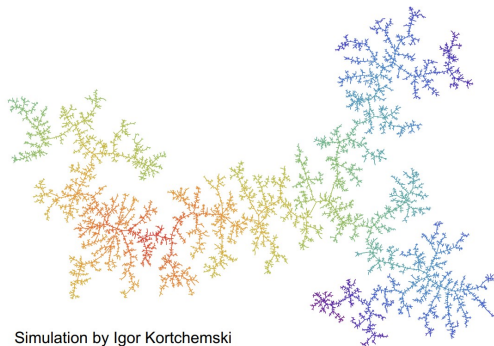
Obtained as the Gromov-Hausdorff scaling limit of Galton-Watson trees (conditioned on n vertices).



Simulation by Igor Kortchemski

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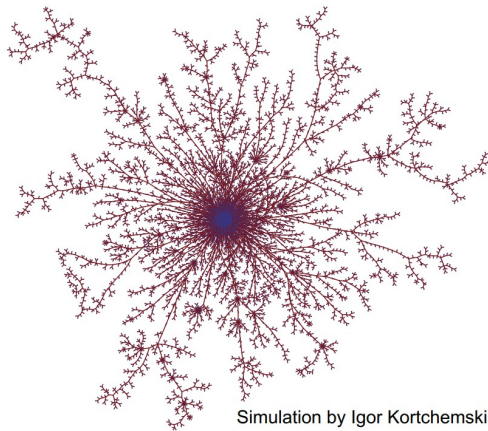
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Gromov-Hausdorff distance defined on isometry classes of (locally)
compact $\mathbb{R}$ -trees!

A not so nice $\mathbb{R}$ -tree

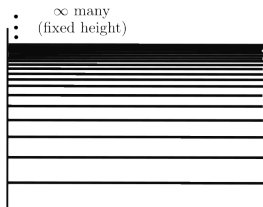
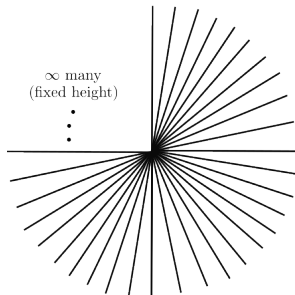
Sometimes you get stuff that starts to look rather nasty in the limit...



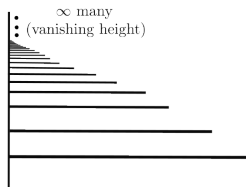
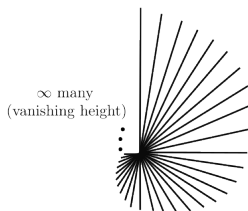
No well-defined Gromov-Hausdorff scaling limit, as any such limit would not be **(locally) compact**.

What does it mean to be locally compact?

Not locally compact



Locally compact



Goal (heuristically): To develop a new ('weaker') **topology**, which allows us to capture scaling limits of $\mathbb{R}$ -trees which are **Polish** but not necessarily **locally compact**!

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⇒ Construction of Lévy forests without Grey's condition!

Mass erasure

Mass-erased subtrees

Let (T, d, ρ) be a Polish rooted $\mathbb{R}$ -tree. Given a point $\sigma \in T$, let $\theta_\sigma T$ be the subtree above σ .

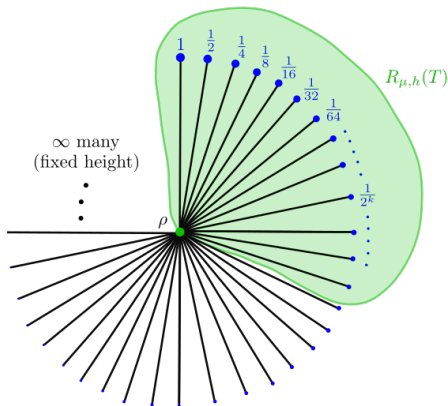
Mass erasure: Equip (T, d, ρ) with a finite Borel measure μ . Given some erasure parameter $h \geq 0$, erase subtrees with mass $< h$. I.e., define the mass-erased subtree

$$R_{\mu,h}(T) := \{\rho\} \cup \{\sigma \in T \mid \mu(\theta_\sigma T) \geq h\}.$$

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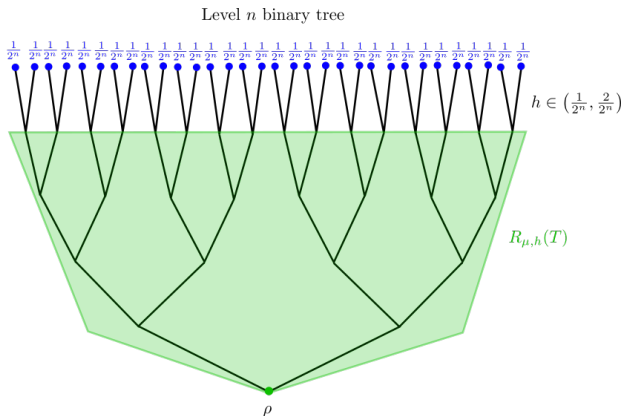
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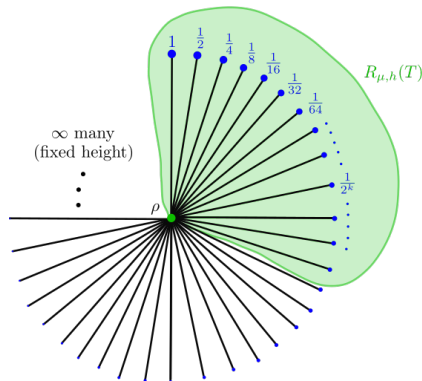
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Mass-erased subtrees

Proposition (Duquesne & Winkel, '26+)

Let (T, d, ρ) be a Polish rooted $\mathbb{R}$ -tree, equipped with a *finite* Borel measure μ . If $h > 0$, then $R_{\mu,h}(T) \subseteq T$ is a *compact discrete* $\mathbb{R}$ -tree. (i.e., $R_{\mu,h}(T)$ is spanned by finitely many points)



Idea of convergence in the sense of mass erasure:

$$d_{\text{GH}}([R_{\mu_n, h}(T_n)], [R_{\mu, h}(T)]) \rightarrow 0 \quad \text{as } n \rightarrow \infty \text{ for all } h > 0?$$

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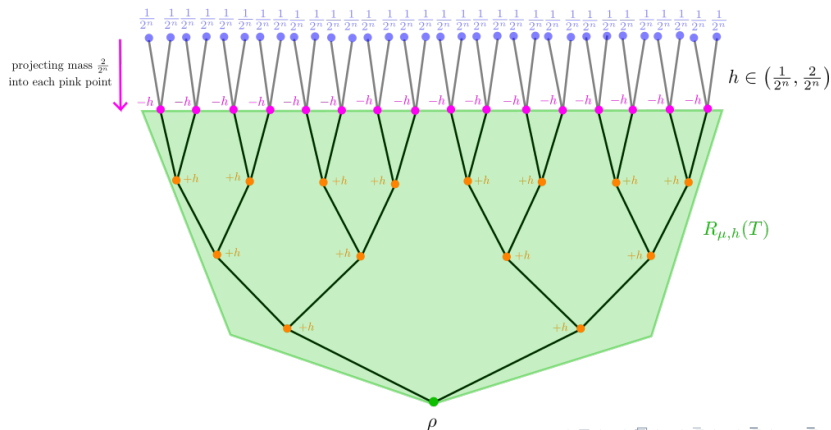
The measures μ_n, μ are intrinsic to this notion, so may want to include information about them in the convergence!

Mass erasure as an operator

Idea: Think of mass erasure as an operator $\mathcal{E}_h : \mathcal{M}_{\text{fin}}(T) \rightarrow \mathcal{M}_{\text{fin}}(T)$ such that

$$\mathcal{E}_h \mu(\theta_\sigma T) = (\mu(\theta_\sigma T) - h)_+ \quad \text{for all } \sigma \in T.$$

$\rightsquigarrow$ There is a unique way to do this on $\mathcal{M}_{\text{fin}}(T)$!



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Definition ((Weak) convergence in the sense of mass erasure)

Let (T, d, ρ) be a **fixed** Polish rooted $\mathbb{R}$ -tree, and let $\mu_n, \mu \in \mathcal{M}_{\text{fin}}(T)$. We say that $\mu_n \xrightarrow{\text{me}} \mu$ if:

$$\forall h > 0 : \quad \mathcal{E}_h \mu_n \xrightarrow{\text{wk}} \mathcal{E}_h \mu \quad \text{as } n \rightarrow \infty.$$

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$\rightsquigarrow$ Induces a metric, which gives rise to a **topology** on $\mathcal{M}_{\text{fin}}(T)$.

Mass erasure as an operator

Theorem (Duquesne & Winkel, '26+)

Let (T, d, ρ) be a Polish rooted $\mathbb{R}$ -tree, and let $\mu_n, \mu \in \mathcal{M}_{fin}(T)$. Then:

$$\mu_n \xrightarrow{wk} \mu \quad \Rightarrow \quad \mu_n \xrightarrow{me} \mu \quad \Rightarrow \quad \forall h > 0 : d_H(R_{\mu_n, h}(T), R_{\mu, h}(T)) \rightarrow 0$$

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Can be extended to the '[Gromov-setting](#)' for GP-isometry classes of finitely measured minimal Polish rooted $\mathbb{R}$ -trees ($\mathbb{T}_{min}^{fin}$).

Theorem (Duquesne & Winkel, '26+)

Let $\mu_n = [T_n, d_n, \rho_n, \mu_n]$ and $\mu = [T, d, \rho, \mu]$ be isometry classes in $\mathbb{T}_{min}^{fin}$. Then:

$$\begin{aligned} \mu_n \xrightarrow{GP} \mu &\Rightarrow \mu_n \xrightarrow{GPme} \mu \\ &\Rightarrow \forall h > 0 : d_{GH}([R_{\mu_n, h}(T_n)], [R_{\mu, h}(T)]) \rightarrow 0 \end{aligned}$$

Theorem (G., '26)

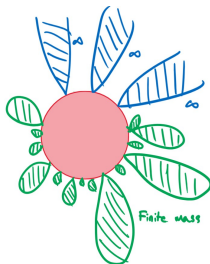
Let (T, d, ρ) be a Polish rooted $\mathbb{R}$ -tree, and let μ_n, μ be *discretely infinite* measures on T . Then

$$\begin{aligned}\mu_n &\xrightarrow{\text{vg}} \mu + \text{'tightness conditions'} \Rightarrow \\ \mu_n &\xrightarrow{\text{vme}} \mu + \text{'tightness conditions'} \Rightarrow \\ \forall h > 0 : d_H^{\text{loc}}(R_{\mu_n, h}(T), R_{\mu, h}(T)) &\rightarrow 0\end{aligned}$$

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Let (T, d, ρ) be a Polish rooted $\mathbb{R}$ -tree, and let μ_n, μ be **discretely infinite** measures on T . Then

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An (infinite) measure μ on (T, d, ρ) is **discretely infinite** if $(\forall r > 0)$:

- ① μ is **boundedly finite**,
- ② $\mu(\theta_\sigma T) = +\infty$ for at most finitely many $\sigma \in \partial B_T(\rho, r)$
- ③ total mass of finite μ -mass subtrees at level r is finite

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Theorem (G., '26+)

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Overview

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- Mass erasure with **finite** measures $\rightsquigarrow$ useful for **bounded** trees [DW26+]

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Applications to Lévy forests

Grimvall's invariance principle

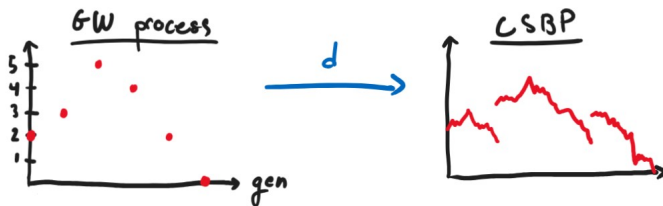
Consider Galton-Watson processes governed by

- an **offspring** distribution ξ on $\mathbb{N}_0$
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Grimvall's invariance principle

Let $N^{(n)} = (N_j^{(n)})_{j \in \mathbb{N}_0}$, $n \in \mathbb{N}$ be a sequence of discrete-time $\text{GW}(\xi_n, \varrho_n)$ processes, with **offspring** distributions ξ_n and **initial** distributions ϱ_n .

- Set $\nu_n(k-1) = \xi_n(k)$ for $k \in \mathbb{N}_0$ (**left-shifted offspring distributions**);
- Let $(\gamma_n)_{n \in \mathbb{N}} \subseteq (0, \infty)$ be a **scaling sequence** with $\gamma_n \rightarrow \infty$.

Theorem (Grimvall, '74)

The following two statements are equivalent:

- ① $\nu_n \left(\frac{\cdot}{n} \right)^* \xrightarrow{wk} \nu$ and $\varrho_n \left(\frac{\cdot}{n} \right) \xrightarrow{wk} \varrho$ as $n \rightarrow \infty$.
- ② The $\text{GW}(\xi_n, \varrho_n)$ -processes have a **scaling limit**, namely

$$\left(\frac{1}{n} N_{\lfloor \gamma_n t \rfloor}^{(n)} \right)_{t \geq 0} \xrightarrow{d} (Z_t)_{t \geq 0} \quad \text{as } n \rightarrow \infty \quad \text{on } \mathcal{D}([0, \infty), \mathbb{R})$$

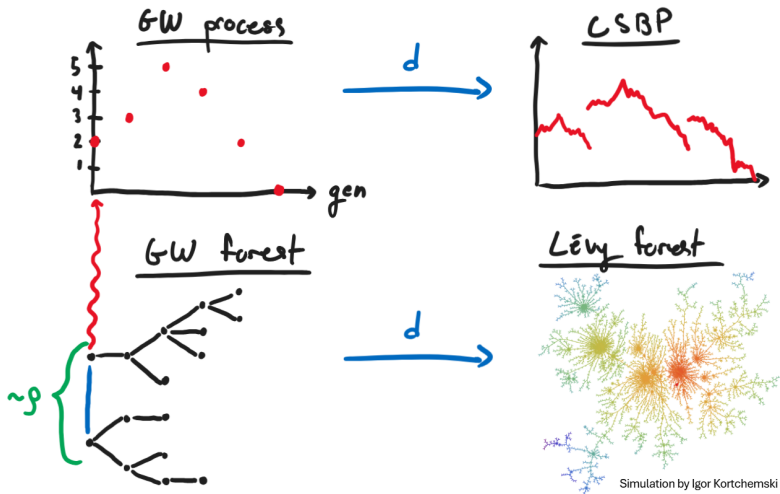
where $Z = (Z_t)_{t \geq 0}$ is a $\text{CSBP}(\psi, \varrho)$ and ψ is the log-Laplace transform of ν .

The big question

Q: Can we obtain a similar invariance principle for the associated genealogical trees?

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Duquesne & Le Gall, '02	Duquesne & Winkel, '19
• (Sub-)critical + Grey • Height processes $\Rightarrow$ GH	• Supercritical + Grey • $\text{GH}^{\text{loc}} \Leftrightarrow$ Leaf-length erasure

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Grey's condition: Let $Z = (Z_t)_{t \geq 0}$ be a $\text{CSBP}(\psi, \varrho)$. Let the **extinction time** be $\tau_{\text{ext}} = \inf\{t \geq 0 \mid Z_t = 0\}$. Then Grey's condition is satisfied if:

$$\mathbb{P}(\tau_{\text{ext}} < \infty) > 0.$$

Equivalently holds, if $\int_0^\infty \frac{1}{\psi(r)} dr < \infty$.

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$\rightsquigarrow$ Essentially ensures that the limiting tree is (locally) compact!

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Invariance principle for (sub-)critical Galton-Watson forests

Theorem (Invariance principle, Duquesne & Winkel, '26+)

Let $(\mathcal{T}_n, d_n, \rho_n)_{n \in \mathbb{N}}$ be a sequence of $GW(\xi_n, \varrho_n)$ $\mathbb{R}$ -forests, satisfying that:

- $\mathbb{E}[\xi_n] \leq 1$ for all $n \in \mathbb{N}$ ((sub-)criticality)
- $\nu_n \left(\frac{\cdot}{n}\right)^{*[n\gamma_n]} \xrightarrow{wk} \nu$ and $\varrho_n \left(\frac{\cdot}{n}\right) \xrightarrow{wk} \varrho$ (Grimvall)

Equip each $\mathbb{R}$ -forest with a measure $\tau_n = \frac{1}{n} \sum_{\sigma \in Lf(\mathcal{T}_n)} \delta_\sigma$. Then:

$$\left(\mathcal{T}_n, \frac{1}{\gamma_n} d_n, \rho_n, \frac{1}{[n\gamma_n] \xi_n(0)} \tau_n \right) \xrightarrow{d} (\mathcal{T}, d, \rho, \tau) =: \tau \quad \text{on } (\mathbb{T}_{min}^{fin}, d_{GPme}),$$

for some limiting (ψ, ϱ) -Lévy forest τ . The convergence holds in $(\mathbb{T}_{min}^{fin}, d_{GP})$ if and only if ψ is the Laplace exponent of a spectrally positive Lévy process with unbounded variation.

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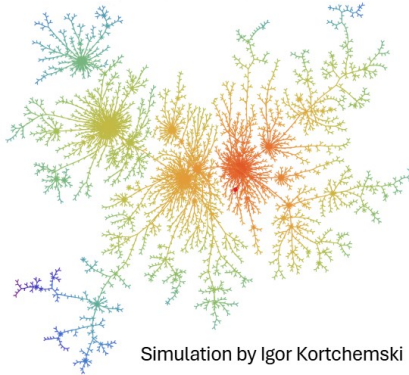
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$\rightsquigarrow$ Working towards extending this to the supercritical case by considering mass erasure with infinite measures!

Thank you for listening!



Tak for opmærksomheden. :)

Main references

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- ⑤ G. (2026+). *Mass erasure of infinitely measured Polish $\mathbb{R}$ -trees I: Construction, continuity, and convergence*. Available soon.
- ⑥ G. (2026+). *Mass erasure of infinitely measured Polish $\mathbb{R}$ -trees II: Applications to supercritical Galton-Watson and Lévy forests*. In preparation.
- ⑦ A. Grimvall (1974). *On the convergence of a sequence of branching processes*. Annals of Probability, Vol. 2(1974).